From Objective Verification to Consumer Perception: A Retrieval-Augmented Contrastive Framework for Misleading-Advertising Detection in Livestream E-Commerce

(Experiments and Conclusion)

4 Experiments

Our empirical study is organized around four research questions. **RQ1** asks whether CLAIMARC discriminates perceived-misleading advertising more accurately than the three baseline families, namely objective fact verification, single-stream text classification, and large language models. **RQ2** asks whether retrieval-augmented contrastive learning (RACL) shapes a representation that tightens same-label claim–evidence pairs and separates pairs carrying opposite perceived-risk labels. **RQ3** asks whether the trained model transfers to unseen categories and streamers, and whether a frozen-encoder retrieval library can absorb new domains without parameter re-training. **RQ4** asks which components, retrieval choices, and hyper-parameters drive the result. Section 4.4 answers RQ1; Section 4.5 answers RQ3; Section 4.6 answers RQ2; and Sections 4.7–4.10 answer RQ4 and characterize the deployment behavior of the model.

4.1 Data and Descriptive Statistics

The corpus covers Chinese livestream e-commerce sessions collected between October 2024 and March 2025. After the §3.2.1 pipeline (per-category attribute normalization, streamer-claim alignment, and three-source product-fact extraction), each instance is a (product, attribute) pair, written (p, a). Two kinds of pairs enter the data. Comment-driven pairs receive a perceived-risk label from aligned consumer comments. Objective-negative pairs cover attributes a streamer asserted but for which no consumer comment aligned; these carry no perceived-misleading signal (y=0) and are weighted by the coverage of product-side evidence. The final dataset contains **4,883** pairs spanning **10** first-level categories, **38** sub-categories, **108** livestream rooms, and **1,532** normalized attributes, of which 2,278 are comment-driven and 2,605 are objective-negative. The positive (misleading) class accounts for **25.6%** of the pairs, so the task is a class-imbalanced screening problem rather than a balanced classification one.

We partition the data 70:10:20 by `room_id`, keeping every pair of a room inside a single split so that no streamer appears in both training and test. Table 1 reports the partition. The three splits hold positive rates of 24.2%, 31.1%, and 29.1%, which are close enough to treat the test distribution as representative. Evidence-source coverage, counting how many of the three product-fact sources fire for a pair, is distributed as 1,483 / 1,826 / 1,118 / 456 pairs for 0 / 1 / 2 / 3 sources (mean 1.11), and the instance-level reliability weight $c_{p,a}$ has mean 0.456, median 0.48, and a bulk in $[0.18, 0.82]$. A streamer-claim segment runs about 23 characters and a product-fact text about 22 characters. Among all pairs, 46.7% align to at least one relevant consumer comment, with 6.9 aligned comments per pair on average and 1.5 of them negative; these aligned comments form the empirical basis of the perceived-risk label. The category counts in Table 2 are long-tailed, led by

Split	#Pairs	Pos. %	#Rooms
Train	3,636	24.2	92
Val	392	31.1	5
Test	855	29.1	11
All	**4,883**	**25.6**	**108**

Table 1: Room-grouped partitions (70:10:20 by `room_id`). Comment-driven pairs: 2,278; objective-negative pairs: 2,605. Evidence-source coverage (0/1/2/3 sources): 1,483 / 1,826 / 1,118 / 456. Reliability weight $c_{p,a}$: mean 0.46, median 0.48, range $[0.18, 0.82]$.

Category	#Pairs	Category	#Pairs
apparel_and_underwear	1,274	smart_home	324
general	827	digital_and_electronics	270
baby_kids_and_pets	715	sports_and_outdoor	262
shoes_and_bags	476	beauty_and_personal_care	225
food_and_beverages	426	jewelry_and_collectibles	84

Table 2: Pair counts across the 10 first-level categories.

apparel, baby-and-kids, and a general bucket, and this tail is what the cross-category transfer study in Section 4.5 stresses.

4.2 Baselines

We compare against three baseline families, each chosen to rule out a competing explanation for the task. Every non-LLM system uses the identical split, the same input budget, the same reliability-weighted supervision, and a threshold selected on the validation set; all such systems were trained on a single 24 GB RTX 4090.

(A) Objective fact verification. This family measures how far one can get by treating perceived-misleadingness as a checkable logical contradiction between the streamer claim and the product evidence, and spans the standard spectrum of pairwise entailment models. **ESIM** (Chen et al. 2017) uses a BiLSTM with soft attention alignment, difference-and-product composition, and pooling; the **Decomposable Attention** model (Parikh et al. 2016) replaces the recurrence with an attend–compare–aggregate scheme, a standard non-recurrent NLI baseline. A Chinese **BERT-NLI cross-encoder** reads the claim and evidence jointly with a pair-classification head. This spectrum measures the discriminative ceiling of the "logical contradiction is mechanically checkable" hypothesis.

(B) Text classification. This is the family closest to our method. The from-scratch neural baselines are **TextCNN** (Kim 2014) and a **BiLSTM** with pooled states; the fine-tuned single-stream encoders **BERT-base-Chinese** and **Chinese-RoBERTa-wwm-ext** are fed `[CLS]` X^c `[SEP]` X^e `[SEP]` and let the `[CLS]` token carry the comparison implicitly, with the whole-word-masked RoBERTa guarding against attributing any gap to backbone capacity.

(C) Frozen-encoder retrieval probes. This family tests what the representation already contains before any contrastive training: logistic regression, a linear SVM, and an MLP on frozen BGE pair features, and a reliability-weighted k-nearest-neighbor voter on frozen BGE embeddings. The k-NN voter reuses CLAIMARC's exact inference rule on an untrained space and is the tightest control on the value of RACL, echoing the retrieval-guided contrastive setup of RGCL (Mei et al.

2024).

(D) Large language models. This family probes how far a general model goes under zero-shot, few-shot, and matched-supervision fine-tuning. We query four gateway models (GPT-5.4, Qwen-Flash, Gemini-3.5-Flash, Kimi-K2.6) under **zero-shot** and **five-shot chain-of-thought (CoT)** prompting, giving each the attribute, the streamer claim, and the product facts in parallel while withholding consumer comments and weak labels. We also fine-tune **Qwen2.5-7B-Instruct with LoRA** on exactly the training pairs and supervision that CLAIMARC uses, which isolates the framework design from raw parametric capacity.

4.3 Evaluation Metrics

Because the positive class is the minority and the deployment goal is to rank claims so a compliance reviewer inspects the riskiest first, we report one set of metrics for every classification experiment, which keeps comparisons across tables consistent. Two are threshold-free. **AP** (average precision, the area under the precision–recall curve, also written AUPRC) is our primary metric, since it is the standard summary of ranking quality when positives are rare. **AUC** (the area under the ROC curve, AUROC) measures threshold-free separability. Two are threshold-dependent: **Accuracy** and the positive-class **F1** ($\mathrm{F1}_{\mathrm{pos}}$), both computed at a single threshold chosen on the validation set and then applied unchanged to the test set. CLAIMARC operates at a high-recall point, which suits screening, so the $\mathrm{F1}_{\mathrm{pos}}$ it reports already balances precision and recall at that point. We do not report Matthews correlation or Macro-F1: the former has no comparable continuous score on the degenerate zero-shot LLM predictions, and the latter is nearly saturated across competent systems and adds little beyond the four metrics above. Pairwise significance uses a paired bootstrap (n=2,000) over the test predictions.

Implementation. The shared backbone is BGE-large-zh-v1.5 (d=1024), fine-tuned end-to-end together with the fusion module and task heads; this places CLAIMARC on the same footing as the fully fine-tuned encoder baselines it is compared against. A parameter-efficient configuration that keeps the backbone frozen and adapts it with LoRA (r=16, α=32) on the query and value projections is reported as an ablation (Section 4.7) and underlies the incremental library-based deployment in Section 4.9; it recovers most of the ranking quality while training a quarter of the parameters. TwoStreamFusion stacks N=2 pre-LN blocks with 8 heads, shared bidirectional cross-attention, and a SwiGLU feed-forward. Training runs in two stages: three warm-up epochs of reliability-weighted binary cross-entropy, then six epochs that add RACL with weight λ_{CL}, rebuilding the per-$(\text{attribute}, y)$ FAISS index at each epoch. We set τ=0.07, K_p=3, K_n=5, and select λ_{CL}=0.5 on the validation set (Section 4.9). Optimization uses AdamW with an encoder learning rate of 1×10^{-5}, a head learning rate of 1×10^{-4}, bf16 precision, and an effective batch of 36. Method-wise, the retrieval-augmented contrast is class-balanced and draws its negatives from the global label pool, the configuration that the ablation in Section 4.7 selects. At inference CLAIMARC reads out its forward classifier (CLS), and the headline classification metrics in every table are taken from it; retrieval-augmented K-nearest-neighbor voting (RKC) does not enter the default prediction but is used where it earns its keep—to absorb new domains without re-training (Section 4.5) and, together with CLS, to gate human abstention in selective prediction (Section 4.10). Every learned system—CLAIMARC and the fine-tuned baselines—is trained under three random seeds (0, 1, and 2), and we report the mean ± standard deviation over these three runs in every in-distribution table (Tables 3 and 7); the deterministic frozen-encoder and language-model baselines are run once. The cross-domain study (Table 4) reports the mean and standard deviation across the held-out folds for the leave-one-category protocol and across three seeds for the leave-20-streamers protocol.

4.4 Main Comparison (RQ1)

Table 3 reports the in-distribution test results across the 14 baselines, as the mean ± standard deviation over three seeds for the trainable systems. CLAIMARC is best on all four metrics—Accuracy 82.6, F1 73.4, AP 75.4, and AUC 90.4—under the same end-to-end fine-tuning budget as the encoder baselines it is compared against. The margin is widest on the two threshold-free ranking metrics the framework is built to optimize: CLAIMARC improves AP by 6.4 points over the best fine-tuned encoder (RoBERTa-CLS, 69.0) and by 7.2 over BERT-CLS. It also edges the strongest baseline on the threshold-point metrics, where BERT-base reads 71.7 F1, so the advantage is not confined to ranking. The precision–recall and ROC curves in Figure 1 make the gap concrete: the separation is largest in the high-recall region above 0.6, which is exactly where a screening system operates.

Reading down the families gives five results. First, objective fact verification is the weakest learned family. ESIM reaches only 51.2 AP, the Decomposable Attention model 60.5, and even a BERT-NLI cross-encoder that reads the claim and evidence jointly stops at 67.9. A claim that consumers later judge misleading rarely contradicts the product evidence in any mechanical sense, so a model built to detect logical contradiction has little to work with. This is direct evidence that perceived-misleadingness is not reducible to claim–evidence inconsistency. Second, the from-scratch neural text classifiers —TextCNN (60.9 AP) and BiLSTM (66.3)—are clearly limited in ranking quality, showing that a sequence model without retrieval pre-training cannot capture the perceived-risk signal. Third, the zero-shot language models fail outright. Their AP sits at 27–29 and their AUC at or below 0.50, no better than random ranking, and five-shot CoT does not move them; Gemini-3.5 and Kimi collapse to a near all-negative prediction, so their nominal accuracy is not informative. Even matched supervision is not sufficient on its own: the LoRA-fine-tuned Qwen2.5-7B closes most of the zero-shot gap (AP 27 → 63.8) yet still trails CLAIMARC by more than 10 AP and trails the much smaller BERT-base, despite carrying roughly 20× CLAIMARC's parameters. Fourth, and most directly, the retrieval machinery alone does not explain the result. A non-parametric control that runs the same reliability-weighted k-NN vote on *frozen* BGE pair embeddings reaches only 64.4 AP—about 11 points below CLAIMARC—and the strongest non-contrastive probe of the frozen space, an MLP, tops out at 67.6. Retrieval voting helps only once contrastive training has reshaped the embedding geometry; the gain comes from the learned representation, not from the inference rule. Fifth, CLAIMARC's advantage is concentrated in ranking and holds up out of distribution: it leads every learned baseline on AP and AUC in distribution and, as Section 4.5 shows, widens that lead on unseen streamers and categories, where the threshold-point metrics of the fine-tuned encoders erode faster.

4.5 Cross-Domain Generalization (RQ3)

We turn the deployment claim into a test. Every system is trained on the source domains and evaluated on an unseen target with no target adaptation; CLAIMARC's retrieval library is built only from the source pairs. Table 4 reports Accuracy, AUC, AP, and positive-class F1 under two protocols. The comparison spans all four paradigms, including the two language-model prompting modes.

Under **leave-one-category** transfer, where each of the 10 first-level categories is held out in turn and the result is averaged over the 10 folds, CLAIMARC is best on Accuracy (81.8), AUC

Group	System	Acc	F1	AP	AUC
(A) Objective fact verification	ESIM	73.8 ±0.6	65.9 ±2.1	51.2 ±1.6	79.9 ±1.3
	Decomposable Attention	76.2 ±0.4	61.5 ±1.6	60.5 ±2.6	83.6 ±0.1
	BERT-NLI cross-encoder	79.1 ±0.5	65.1 ±3.2	67.9 ±1.1	87.4 ±0.6
(B) Text classification	TextCNN	76.5 ±0.8	57.0 ±2.6	60.9 ±0.9	84.1 ±0.8
	BiLSTM	79.2 ±0.3	65.5 ±2.6	66.3 ±0.9	86.4 ±0.3
	BERT-base + [CLS]	80.9 ±0.5	71.7 ±1.0	68.2 ±1.8	88.3 ±0.4
	RoBERTa-wwm + [CLS]	80.5 ±0.8	70.0 ±1.8	69.0 ±1.0	88.4 ±0.1
(C) Frozen-encoder retrieval probes	BGE-frozen + LR	77.0	68.5	66.4	86.6
	BGE-frozen + linear SVM	76.0	68.0	60.5	84.8
	BGE-frozen + MLP	78.6	67.4	67.6	86.4
	BGE-frozen + kNN vote	78.1	62.1	64.4	83.0
(D) Large language models	Qwen-Flash (0-shot)	49.8	30.5	27.4	46.1
	GPT-5.4 (5-shot CoT)	58.2	24.8	26.9	44.6
	Qwen2.5-7B + LoRA SFT	73.0	61.7	63.8	81.9
CLAIMARC	(ours)	**82.6** ±0.4	**73.4** ±0.5	**75.4** ±1.3	**90.4** ±0.4

Table 3: Main comparison on the test set (%, N=855, 249 positives) across 14 baselines in four groups. Trainable systems (ESIM, Decomposable Attention, BERT-NLI, TextCNN, BiLSTM, the fine-tuned encoders, and CLAIMARC) report mean ± standard deviation over three seeds; the frozen-encoder probes and the language models are run once. Accuracy and F1 use a validation-selected threshold; AP and AUC are the primary threshold-free metrics. Group (C) applies linear, MLP, and reliability-weighted kNN heads to *untrained* BGE pair embeddings, isolating what the representation already encodes before any contrastive training. Representative LLM rows are shown; all gateway models in zero-shot/5-shot score AP 27–29 and AUC $\leq$ 0.50, and Gemini-3.5 and Kimi collapse to a near all-negative prediction, so their accuracy is not informative. Best per column among the learned systems in bold.

(90.0), and AP (71.0), and holds the smallest spread across categories (AUC standard deviation 4.0 against BERT-CLS's 4.7). The best fine-tuned competitor, BERT-CLS, reads 89.4 AUC and 70.1 AP; CLAIMARC keeps a one-point ranking edge while transferring through a source-only library, so the advantage comes from the structural prior rather than from re-fitting parameters to the target. Under **leave-20-streamers** transfer, where a fixed set of 20 comment-volume-stratified rooms is held out and the result is averaged over three seeds, CLAIMARC leads both ranking metrics by a clear margin—AUC 93.1 and AP 81.4, against BERT-CLS (92.2/78.7) and RoBERTa-CLS (92.0/78.9)—and stays within seed noise of the best encoder on the threshold-point metrics (Accuracy 84.2 versus RoBERTa-CLS's 84.5; F1 73.4). The AP lead of roughly 2.5 points over the strongest encoder is wider than in distribution, which says the attribute structure CLAIMARC encodes carries across streamers even when speaking style does not. Under both protocols the zero-shot and five-shot language models stay near or below random (AUC $\leq$ 0.52), so a general model's parametric prior does not transfer to the consumer-perception axis. The consistent thread is that CLAIMARC's edge is concentrated in ranking quality (AP, AUC) and survives both forms of distribution shift, which is exactly the regime a screening tool faces.

The comparison above freezes every system and reads it once on the target. The framework, however, claims something stronger: that a *frozen* CLAIMARC can keep improving on a new domain by maintenance of its retrieval index alone, with no gradient step. We test this directly under streamer transfer. The encoder is trained on the source rooms and then frozen; we then write a growing fraction of the held-out streamers' labelled pairs into the retrieval library and re-read the retrieval vote on the remaining held-out streamers, which are never written. The forward classifier

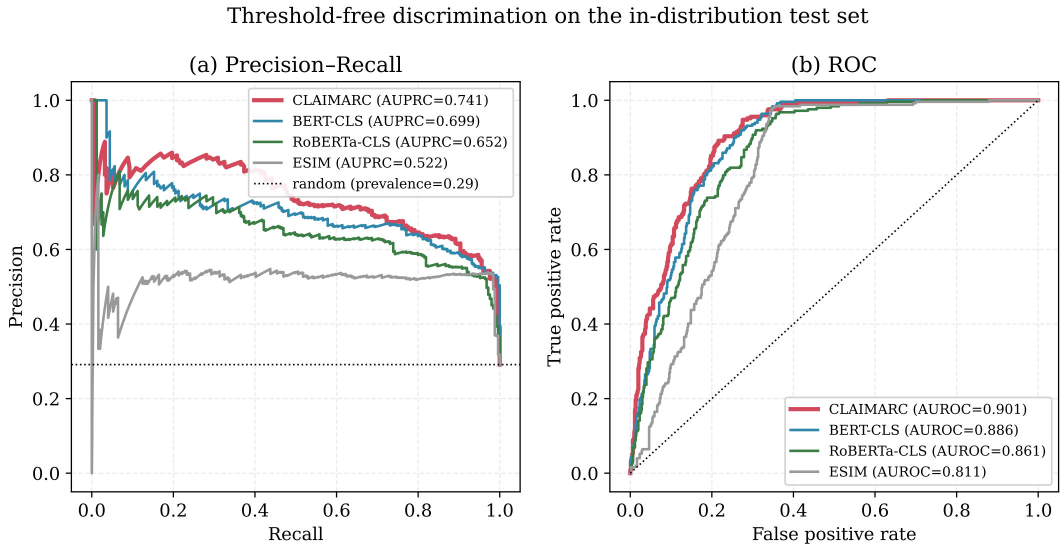

Figure 1: Threshold-free discrimination on the test set. CLAIMARC (red) leads the precision–recall curve, most visibly in the high-recall screening region, and the ROC curve. The dotted line in (a) marks the positive prevalence (0.29).

is fixed throughout—being parametric, it cannot absorb the new rooms without re-training—so it serves as the no-adaptation reference. Table 5 traces the result. The retrieval vote, the one component that can adapt, climbs monotonically as the index fills: AP rises from 65.5 with a source-only library to 71.5 at full target coverage (+6.0), AUC from 80.3 to 84.8 (+4.5), and the operating-point F1 from 72.4 to 74.9, overtaking the frozen forward classifier's F1 (73.5) once the library is filled. Crucially the gain is smooth and shows up early—half of it is realized once only a fifth of the target pool is in the index—so a deployment can absorb a new streamer or a new risk phrasing incrementally, at the cost of one encode and a write per pair. This is the concrete mechanism behind the deployment argument: because RACL has shaped the embedding space so that perceived-risk structure is locally consistent, dropping a handful of target exemplars into the neighborhood is enough to re-point the vote, whereas a parametric model would need a fresh training pass to move at all.

4.6 Representation Geometry and Boundary Sensitivity (RQ2)

The value of a contrastive objective should show up in how it reshapes the label geometry, making misleading and honest pairs more separable in the retrieval space—the structure the RKC vote (Section 4.5) and the gradient-free library write read off. To separate this effect from prior work we run a strictly controlled comparison: an identical BGE-full-fine-tuning backbone, classifier head and hyper-parameters, with three variants that differ *only* in the contrastive objective—*no contrast* (BCE only), standard supervised contrastive learning (*SupCon*; Khosla et al. 2020, with in-batch same-label positives and other-label negatives, no retrieval or hard mining), and our retrieval-augmented contrast (*RACL*; full-pool pseudo-gold positives and hard negatives, reliability- and class-weighted). All metrics are measured on the test embeddings $\mathbf{g}_{p,a}$ over three seeds and are attribute-free (the train split serves as the retrieval index, the test split as queries); Table 6 reports them.

Three findings emerge. First, **class separability**: RACL reaches a label silhouette of 0.422,

System	Acc	AUC	AP	F1
(a) Leave-one-category (mean±s.d. over 10 folds)				
ESIM	79.3 ±7.2	85.1 ±6.7	56.5 ±12.2	**70.1** ±8.6
BERT-CLS	81.0 ±6.5	89.4 ±4.7	70.1 ±8.0	67.5 ±5.0
RoBERTa-CLS	81.0 ±4.6	88.5 ±4.3	67.3 ±6.7	66.2 ±8.1
LLM zero-shot	49.6 ±3.8	45.2 ±5.1	25.6 ±12.6	29.0 ±13.1
LLM 5-shot CoT	46.0 ±4.8	45.6 ±4.5	25.7 ±12.1	30.6 ±12.6
CLAIMARC	**81.8** ±5.0	**90.0** ±4.0	**71.0** ±5.5	66.8 ±5.2
(b) Leave-20-streamers (mean±s.d. over 3 seeds)				
ESIM	82.9 ±1.0	88.2 ±0.4	68.4 ±0.7	**77.0** ±1.9
BERT-CLS	83.6 ±1.3	92.2 ±0.0	78.7 ±0.3	73.3 ±5.2
RoBERTa-CLS	**84.5** ±0.4	92.0 ±0.2	78.9 ±0.8	75.6 ±1.4
LLM zero-shot	49.0	47.0	29.3	31.7
LLM 5-shot CoT	48.8	52.1	32.4	43.3
CLAIMARC	84.2 ±1.3	**93.1** ±0.3	**81.4** ±1.3	73.4 ±3.7

Table 4: Cross-domain generalization (%) under two protocols, reporting Accuracy, AUC, AP, and positive-class F1. All systems train on the source and test on an unseen target with no target adaptation; CLAIMARC's retrieval library is built only from the source pairs. Panel (a) reports the mean ± standard deviation across the 10 held-out first-level categories; panel (b) reports the mean ± standard deviation across three seeds for a fixed set of 20 held-out, comment-volume-stratified rooms. The learned systems are compared with the two language-model prompting modes (run once). CLAIMARC leads on the ranking metrics (AP, AUC) under both protocols and holds the tightest category spread (AUC 90.0 ± 4.0 versus BERT-CLS 89.4 ± 4.7); under streamer transfer it is within seed noise of the best encoder on Accuracy and F1 while leading AP by ∼2.5 points. The in-domain reference AP is 75.4 (Section 4.4). Best per column among the learned systems within each panel in bold.

above standard SupCon (0.349) and no-contrast (0.196), and with the smallest variance (±0.010 vs. SupCon's ±0.173)—the most stable separation of the three. The UMAP projection in Figure 3 shows this by eye: RACL condenses the misleading pairs into a detached cluster, the two classes interleave without contrast, and standard SupCon sits in between. Second, the **key difference between RACL and standard SupCon is whether the representation collapses**: SupCon attains the tightest same-label alignment (0.478) but at the cost of uniformity (only −0.86), squeezing same-class points together; RACL instead keeps the space spread out (uniformity −1.21) while separating the classes, landing in the retrieval-friendlier regime of the alignment–uniformity plane and avoiding the collapse of the vanilla objective (Wang and Isola 2020). Third, the **hard region**: restricted to the confusable subset (queries whose nearest opposite-label neighbor is most similar), both contrastive objectives lift neighborhood purity from 0.50 to 0.57—the contrastive signal acts precisely on pairs that look alike yet carry opposite labels, which is where RACL applies explicit pressure through hard negatives. We note that neighborhood purity@10 is comparable across the three (≈0.81): full fine-tuning already establishes label-consistent neighborhoods, so the distinct contribution of the contrastive objective is class *separation*, not raw neighbor purity. Taken together, contrastive training yields a more separable yet uncollapsed retrieval geometry—the structure that the retrieval vote and library-write adaptation depend on, and the reason the frozen-encoder vote becomes useful only after the representation is reshaped (Section 4.5).

Retrieval library on held-out streamers	AP	AUC	F1
Frozen forward classifier (no adaptation)	81.1 ±0.9	93.1 ±0.3	73.5 ±2.9
source pairs only	65.5 ±1.8	80.3 ±2.6	72.4 ±1.9
+ 20% of target pool	67.5 ±1.1	81.3 ±2.2	73.3 ±1.9
+ 40%	69.0 ±1.2	82.7 ±1.7	73.8 ±2.6
+ 60%	69.7 ±1.0	83.3 ±1.8	73.6 ±2.4
+ 80%	70.7 ±1.2	84.3 ±1.6	74.4 ±2.6
+ 100% of target pool	**71.5** ±1.5	**84.8** ±1.5	**74.9** ±2.5

Table 5: Gradient-free library adaptation under leave-20-streamers transfer (%, mean ± s.d. over 3 seeds). The encoder is frozen after source training; a growing fraction of the held-out streamers' labelled pairs is written into the retrieval library, and the retrieval vote is read on the disjoint, never-written half of the held-out streamers. The vote—the only adaptable component—improves monotonically with library coverage and overtakes the frozen forward classifier on F1 at full coverage, without any gradient step.

Variant	Silh. ↑	Hard pur.@10 ↑	Align. ↓	Unif.	Pur.@10
w/o contrast	0.196±0.053	0.500±0.043	1.121	−2.02	0.811
SupCon (Khosla et al. 2020)	0.349±0.173	**0.573**±**0.017**	0.478	−0.86	0.804
RACL (ours)	**0.422**±**0.010**	**0.573**±**0.088**	0.905	−1.21	0.811

Table 6: Label-conditional geometry of the test retrieval embedding under three contrastive objectives that share an identical BGE-full-fine-tuning backbone, classifier and hyper-parameters (3-seed mean±s.d.; attribute-free, with the train split as the retrieval index). RACL attains the best and most stable label silhouette; standard SupCon reaches the tightest alignment but a collapsed (least uniform) space, whereas RACL preserves uniformity while separating the classes. Neighborhood purity@10 is already established by full fine-tuning and is comparable across objectives.

4.7 Ablation Studies (RQ4)

We dissect the framework one design choice at a time on the fully fine-tuned canonical, holding everything else fixed and scoring the forward classifier so that all variants sit on the same footing. We follow the organization of recent retrieval-contrastive work (Mei et al. 2024) and report three focused tables, each isolating one family of choices: the core objective, weighting, and backbone components (Table 7), the dual-stream architecture (Table 8), and the retrieval-augmented contrast's positive/negative mining (Table 9). All numbers are mean ± standard deviation over three seeds; the canonical is the bold reference in each table.

Core objective, weighting, and backbone components. Table 7 removes one component at a time, and every change costs both the operating point and the ranking metric. Dropping the retrieval-augmented contrast (cross-entropy only) costs 2.6 F1 (73.4→70.8) and 3.3 AP; removing the reliability weight that down-weights noisy labels costs 1.8 F1 and 3.2 AP; and turning off the class-balanced scaling of the contrast—the imbalance-aware refinement that keeps the 26% misleading class from being swamped—costs 0.7 F1 and 1.5 AP. Collapsing the classifier and retrieval head's ESIM-style four-tuple $[\mathbf{h}_c, \mathbf{h}_e, \mathbf{h}_c - \mathbf{h}_e, \mathbf{h}_c \odot \mathbf{h}_e]$ to a plain concatenation $[\mathbf{h}_c, \mathbf{h}_e]$ costs 3.6 AP and 1.4 F1, so the explicit difference and product features—the discrepancy signal between a claim and its evidence—are not redundant with what the fusion carries. Swapping the retrieval-pretrained BGE for BERT-base costs 4.3 AP, confirming that the ranking draws on the backbone's retrieval pre-training rather than raw encoder size. The contrast's full value is clearest where a head can-

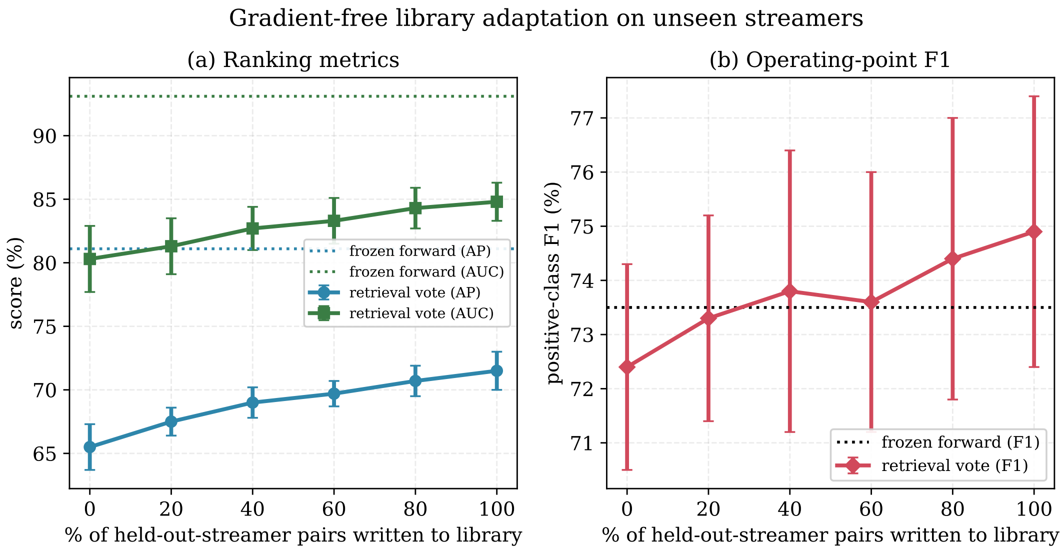

Figure 2: Gradient-free library adaptation (leave-20-streamers, 3-seed mean±s.d.). As held-out-streamer pairs are written into the frozen-encoder retrieval library, the retrieval vote's AP and AUC climb monotonically toward the frozen forward classifier, demonstrating that new streamers are absorbed by index maintenance rather than re-training.

not compensate: the non-parametric probe in Table 3—a reliability-weighted vote on *frozen* BGE pair embeddings—reaches only 64.4 AP, against 75.4 for the contrastively trained encoder, and the cross-domain library study in Section 4.5 shows the same vote becoming useful only after the contrast has shaped the space.

Why two contrasting streams. Table 8 probes the architecture that gives the framework its name. We compare the canonical dual-stream model against three reductions: keeping both streams but removing the cross-attention TwoStreamFusion (concatenation only), and two genuine single-stream models that feed only the claim or only the evidence into the encoder with no fusion at all. The evidence-only model *collapses*—AP falls to 48.4 and AUC to 68.7, barely above the 29.1% prevalence—because the perceived-risk label lives in whether the evidence *contradicts the claim*, a relation that is unreadable once the claim is removed. The claim-only model fares better but still loses 5.0 AP, since without the evidence it cannot tell an honest claim from an unsupported one. With both streams present, removing the fusion still costs 4.6 F1 and 1.9 Accuracy—the largest operating-point drop among all variants—confirming that the cross-stream interaction, not merely the presence of two encoders, is what aligns a claim with its evidence. The two streams and their interaction are jointly necessary.

Retrieval-augmented contrast: which examples to draw. Table 9 ablates the positive and negative mining of the contrast, following the protocol of RGCL (Mei et al. 2024). Two choices that the literature warns about are confirmed here. First, replacing the pseudo-gold positives (highest-similarity same-label neighbors) with hard positives (lowest-similarity same-label neighbors) does not help and slightly lowers F1 (73.0 vs 73.4), matching the report that hard positives destabilize retrieval-guided contrast. Second, restricting negatives—either to same-attribute pairs or to the same evidence type—never beats the simpler class-balanced global pool, so the canonical keeps the global pool. The number of pseudo-gold positives is inert (Kp $\in \{1, 3, 5\}$ all within noise), but

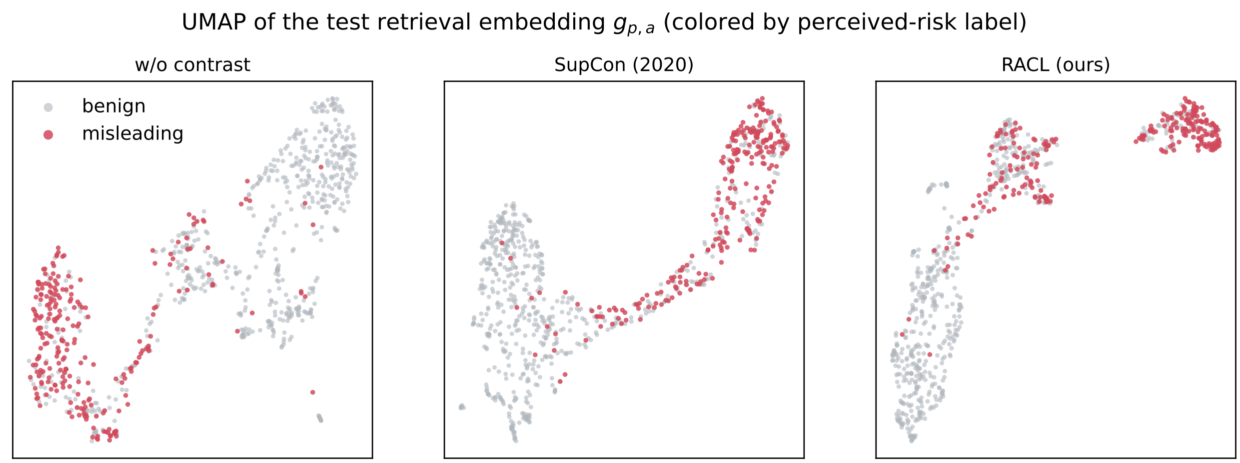

Figure 3: UMAP of the test retrieval embedding $\mathbf{g}_{p,a}$, colored by perceived-risk label, under the three contrastive objectives (w/o contrast | SupCon 2020 | RACL). RACL isolates the misleading pairs into a detached cluster; without contrast the two classes interleave, while standard SupCon lies in between.

Variant (CLAIMARC, forward)	Acc	F1	AP	AUC
Canonical CLAIMARC	**82.6** ±0.4	**73.4** ±0.5	**75.4** ±1.3	**90.4** ±0.4
w/o RACL contrast (CE only)	81.2 ±1.1	70.8 ±2.0	72.1 ±2.3	89.4 ±0.9
w/o reliability weight c	81.6 ±1.3	71.6 ±1.6	72.2 ±3.1	89.5 ±0.7
w/o class-balanced contrast	82.1 ±0.5	72.7 ±1.9	73.9 ±0.4	89.9 ±0.2
w/o ESIM 4-tuple head (concat only)	81.4 ±0.6	72.0 ±1.7	71.8 ±2.3	89.4 ±0.5
backbone: BGE $\rightarrow$ BERT	81.2 ±0.6	72.4 ±0.9	71.1 ±1.4	89.2 ±0.4

Table 7: Core objective, weighting, and backbone components, removed one at a time (%, forward classifier, mean ± s.d. over three seeds; canonical in bold). Every removal lowers both the operating point and the ranking metric, locating the contribution of each component.

the number of hard negatives has a clear optimum: too few (Kn= 1) costs 1.3 F1 and too many (Kn= 10) costs 4.2 F1, so the canonical Kn= 5 is the sweet spot, echoing the finding that piling on hard negatives degrades retrieval-contrastive training.

4.8 Does the Reliability Weight Earn Its Form? (RQ4)

The reliability weight $c_{p,a}$ is one of the components that moves the headline in Table 7, so we ask a sharper question than "does it help": is its *specific* construction warranted, or would any down-weighting of low-signal pairs do? We answer with a controlled study that holds the architecture and the data fixed and perturbs only the per-sample training weight $w_{p,a}$, leaving the validation- and test-set weights untouched so the threshold and the weighted metrics stay comparable across rows. Table 10 reports the forward classifier under seven counterfactual weightings. The reading is consistent. Removing the weight (uniform) costs 2.2 F1, in line with the ablation. *Reversing* it—up-weighting the pairs the design judges least reliable—costs 5.7 F1 and lands below uniform, so the direction of the weight, not merely its presence, carries information: a high c does mark a more trustworthy label. Randomly *permuting* c across pairs, which keeps its marginal distribution but destroys the pair-to-weight correspondence, falls back to the uniform level, so the value is in *which* pairs are up-weighted, not in the shape of the weight histogram. Coarsening c to a median

Stream design	Acc	F1	AP	AUC
Dual-stream + TwoStreamFusion	**82.6** ±0.4	**73.4** ±0.5	**75.4** ±1.3	**90.4** ±0.4
Dual-stream, w/o fusion (concat)	80.7 ±0.5	68.8 ±3.2	73.8 ±1.4	89.6 ±0.4
Claim stream only (no fusion)	81.1 ±1.0	73.0 ±0.3	70.4 ±1.9	89.2 ±0.1
Evidence stream only (no fusion)	68.7 ±0.8	44.1 ±4.3	48.4 ±0.4	68.7 ±1.2

Table 8: Necessity of two contrasting streams (%, forward classifier, mean ± s.d. over three seeds; canonical in bold). A single stream cannot read the claim–evidence relation: the evidence-only model collapses to near-prevalence AP, and even with both streams the cross-attention fusion is needed for the operating point.

RACL retrieval design	Acc	F1	AP	AUC
Pseudo-gold pos., global hard neg. (Kp3/Kn5)	**82.6** ±0.4	**73.4** ±0.5	**75.4** ±1.3	**90.4** ±0.4
Positive / negative type				
hard positives (vs pseudo-gold)	81.9 ±1.1	73.0 ±0.7	73.7 ±0.7	90.0 ±0.2
same-attribute negatives	82.1 ±0.4	72.6 ±1.3	73.5 ±0.5	90.0 ±0.0
same-evidence-type negatives	81.8 ±0.3	72.7 ±1.5	74.0 ±0.5	90.0 ±0.1
Number of retrieved examples				
positives Kp= 1	82.5 ±0.7	73.4 ±0.6	74.0 ±0.7	89.9 ±0.2
positives Kp= 5	82.4 ±0.5	73.4 ±0.2	73.7 ±0.5	90.0 ±0.1
hard negatives Kn= 1	81.4 ±0.4	72.1 ±1.4	73.6 ±0.2	90.0 ±0.1
hard negatives Kn= 10	81.1 ±0.3	69.2 ±2.3	73.5 ±0.4	89.9 ±0.1

Table 9: Retrieval-augmented contrast: positive/negative mining, in the protocol of RGCL (Mei et al. 2024) (%, forward classifier, mean ± s.d. over three seeds; canonical in bold). Pseudo-gold positives beat hard positives, restricting negatives never helps, and the hard-negative count has a clear optimum at five.

split (binary) is the counterfactual that moves the ranking metrics most, dropping AP by 7.0, which says the continuous grading—not just a reliable/unreliable dichotomy—is what protects the precision–recall ordering. Collapsing the composite to its saturation term alone (count) costs 1.5 F1, so the asymmetric-diagnosticity and authenticity factors contribute beyond a raw comment count, and rolling back to the pre-refinement label weights (old c) costs 3.4 F1, so the label-reconstruction iteration was not cosmetic. The lone neutral perturbation is a square-root reshaping, which preserves the ordering and matches the canonical within noise—the design is robust to how steeply it grades reliability but highly sensitive to reversing, flattening, or shuffling it. Two patterns cut across the table: the operating-point metrics (Accuracy, F1) consistently favor the canonical weight while the threshold-free metrics barely move except under binarization, which locates the mechanism precisely—reliability weighting sharpens the decision boundary rather than re-ranking the test set.

4.9 Hyper-parameter Sensitivity and Efficiency (RQ4)

Figure 4 sweeps five core knobs: the number of fusion blocks N, the number of attention heads, the LoRA rank, the contrastive temperature τ, and the retrieval sizes (K_p, K_n); Table 12 in the appendix lists these together with the loss-function, feed-forward, cross-attention, and projection variants, for 26 single-seed configurations in all; to keep the sweep affordable these runs use the parameter-efficient LoRA-frozen variant. AP stays inside a narrow 0.69–0.74 band across the five core knobs. The canonical objective and retrieval settings (λ_{CL}=0.5, τ=0.07, N=2,

Training weight $w_{p,a}$	**Acc**	**F1**	**AP**	**AUC**
Canonical (w=c, the design)	**82.6** ±0.4	73.4 ±0.5	**75.4** ±1.3	**90.4** ±0.4
Uniform (w=1)	81.6 ±1.2	71.2 ±1.6	73.3 ±2.3	89.6 ±0.6
Reversed	80.0 ±0.4	67.7 ±3.5	74.1 ±1.9	89.5 ±0.3
Permuted across pairs	80.4 ±0.6	71.6 ±1.5	73.0 ±2.8	89.4 ±0.8
Median-split (binary)	79.5 ±0.8	70.2 ±2.5	68.4 ±1.1	88.3 ±0.7
Saturation term only	80.3 ±0.3	71.9 ±0.4	73.1 ±1.6	89.1 ±0.4
Pre-refinement (old c)	80.4 ±1.3	70.0 ±3.4	72.1 ±0.8	89.4 ±0.6
Square-root (w=$\sqrt{c}$)	81.8 ±1.5	**74.0** ±1.3	72.6 ±2.5	89.9 ±0.5

Table 10: Counterfactual reliability weightings (%, forward classifier, mean ± s.d. over three seeds). Only the per-sample *training* weight is altered; validation/test weights and the architecture are held fixed. The canonical weight w=c is best on Accuracy, AP, and AUC, with the square-root reshaping tying it on F1; the threshold-free metrics are insensitive to loss re-weighting except under the coarse median split, which breaks the precision–recall ordering (AP −7.0). Reversing the weight is worse than removing it, and permuting it reverts to the uniform level—the design's value is sample-specific and directional, not a generic down-weighting.

System	**Total**	**Trainable**	**New-domain adaptation**
BERT-base (FT)	102M	102M (100%)	gradient retrain
Qwen2.5-7B+LoRA	7,616M	∼20M	LoRA retrain
CLAIMARC (full)	396.5M	396.5M (100%)	gradient retrain
CLAIMARC (LoRA-frozen)	396.5M	**92.7**M (23.4%)	**library write**

Table 11: Deployment cost. The canonical fully fine-tunes the encoder; the parameter-efficient variant trains a minority of its parameters and absorbs new domains by a library write (one 36 ms encode, no gradient step) rather than by retraining, for about a five-point AP trade-off (Section 4.7).

(K_p, K_n)=$(3, 5)$) sit on the flat part of each curve, and only an oversized retrieval set $(5, 10)$ degrades the score by pulling in noisy neighbors. The model is not sensitive to these choices within a reasonable range, which is why the configuration was fixed once on the validation set rather than tuned per metric.

Table 11 compares deployment costs. The canonical CLAIMARC fully fine-tunes its 396.5M parameters and finishes the two-stage schedule in about 18 minutes on one RTX 4090; at inference it encodes a pair in 36 ms and runs RKC retrieval over the 3,636-item library in 0.18 ms per query. The parameter-efficient variant trains only 92.7M parameters (23.4%) with the backbone frozen, trading about five AP points (Table 7) for a decisive operational property recorded in the last column: adapting to a new streamer, category, or risk phrasing costs one 36 ms encode and a write to the library, with no gradient step, whereas every parametric baseline—and the fully fine-tuned canonical itself—needs a retraining pass, and the 7B model also carries about 20× the parameters and an autoregressive decoding cost. Moving the adaptable component out of the model weights and into an external index is what buys this.

4.10 Selective Prediction and Error Analysis

A screening tool runs alongside human reviewers, so its value depends not only on average accuracy but on whether it can flag its own unreliable calls for escalation. CLAIMARC offers a training-free gate for this. The forward classifier and the RKC retrieval vote are two near-independent views of

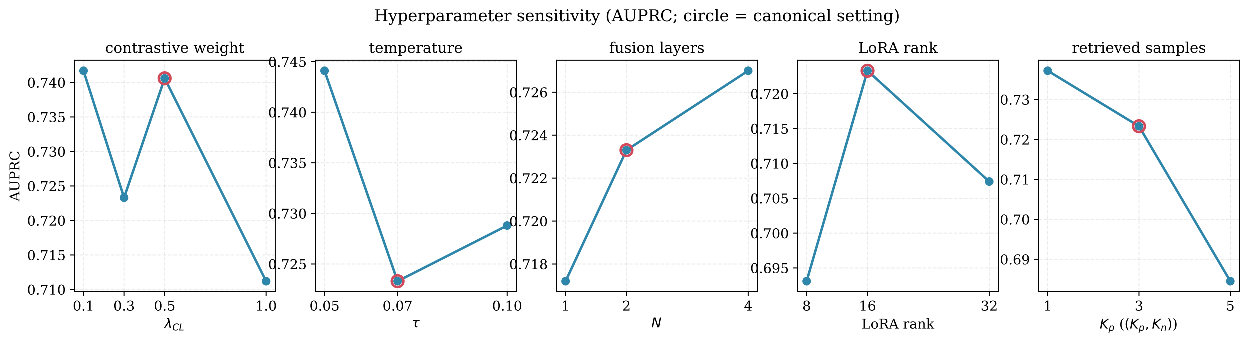

Figure 4: Hyper-parameter sensitivity (AP). Across all five sweeps the score stays inside a narrow band; the canonical configuration (circled) sits on the stable plateau.

the same pair, and their disagreement $|p_{\text{fwd}} - p_{\text{RKC}}|$ (the self-consistency signal of §3.2.9) marks the cases to abstain on and route to a reviewer. Figure 5 traces the risk–coverage trade-off as we abstain on the most-disagreeing pairs, averaging over the three canonical seeds. The auto-decided subset improves monotonically: lowering coverage from 100% to 80%, by escalating the 20% of pairs with the largest disagreement, raises Accuracy from 0.814 to 0.867 and selective AP from 0.737 to 0.778; at 70% coverage Accuracy reaches 0.889 and AP 0.809, and at 65% coverage Accuracy 0.908 and AP 0.835. A random abstention of the same size leaves every metric flat (Accuracy around 0.81, AP around 0.73), so the gate is isolating genuinely hard cases rather than just shrinking the set. In effect the model becomes a calibrated triage front-end that clears the confident majority and surfaces a small, high-yield queue for human review. The reliability diagram in Figure 6 adds that a single temperature scale brings the predicted probabilities close to the diagonal, which supports their use in a decision-support setting.

Turning to the errors, at the validation-selected operating point CLAIMARC recovers 84% of the perceived-misleading pairs (TP=210, FN=39), trading some precision (FP=120) as a screening tool should. The errors concentrate where product evidence is thin: the error rate falls from 24.1% to 16.4% to 10.3% as evidence-source coverage goes from one to two to three sources, which both supports the three-source product-fact channel and points to evidence acquisition, rather than the discriminator, as the limit in the residual regime. Two interpretable error types stand out. False negatives are mostly terse stylized endorsements such as “加绒的”, “质感真的很顶”, and “版型好看” that consumers later judged misleading but that carry no surface contradiction with the product facts; these are the cases the fact-verification and LLM baselines also miss, and the limit here is signal sparsity rather than geometry. False positives concentrate on hyperbolic metaphor over a benign fact, for example “喷上去之后就跟镀了膜似的” for a foam-type product, where the model over-attributes risk to vivid figurative language. Finally, CLAIMARC and BERT make different mistakes, each correcting about 44 of the other’s errors and agreeing on only 114, which points to complementary inductive biases and makes a retrieval-augmented ensemble a natural next step.

5 Conclusion

5.1 Findings

The four research questions resolve as follows. On RQ1, CLAIMARC leads on all four metrics—Accuracy 82.6, F1_{pos} 73.4, AP 75.4, and AUC 90.4—under the same end-to-end fine-tuning budget

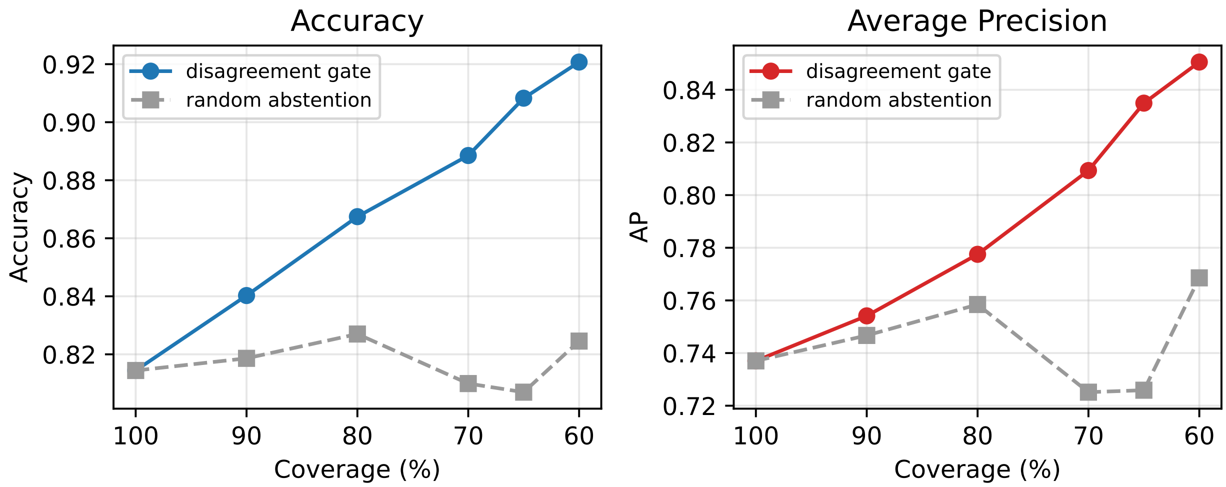

Figure 5: Selective prediction on the test set. Abstaining on the highest forward–RKC disagreement and routing those pairs to human review improves both selective AP and Accuracy on the auto-decided remainder, whereas a random abstention of the same size (dashed) stays flat.

as the encoder baselines, with the widest margins on the threshold-free ranking metrics the framework targets (AP +6.4 over the best fine-tuned encoder). The two competing explanations both fail across the 14 baselines: objective fact verification is the weakest learned family (ESIM AP 51.2, a BERT-NLI cross-encoder 67.9), and the zero-shot LLMs sit at or below random (AP 27–29), with five-shot CoT no better; even a LoRA-fine-tuned Qwen2.5-7B trails by more than 10 AP at roughly 20× the parameters, and a non-parametric probe on frozen embeddings reaches only 64.4 AP—it is the contrastive geometry, not the retrieval rule, that carries the result. On RQ2, under a controlled comparison against standard supervised contrastive learning (SupCon; Khosla et al. 2020), RACL produces the best and most stable class separation (label silhouette 0.422 vs. 0.349 for SupCon and 0.196 without contrast) and lifts hard-region neighbor purity from 0.50 to 0.57; unlike standard SupCon it separates the classes without collapsing the representation (a balanced alignment–uniformity profile), the geometric structure the retrieval vote and library write exploit. On RQ3, CLAIMARC transfers best of all learned systems, leading the ranking metrics on unseen categories (AUC 90.0, AP 71.0) and on unseen streamers (AUC 93.1, AP 81.4, the largest streamer-transfer AP of any system), while the LLMs do not transfer at all; freezing the encoder and writing held-out streamer pairs into the retrieval library then lifts the gradient-free vote monotonically from 65.5 to 71.5 AP, narrowing the gap to the trained forward classifier without a single gradient step. On RQ4, three focused ablation tables—core components and backbone, dual-stream architecture, and retrieval design—each support the design: removing fusion costs 4.6 F1, removing the contrastive objective 2.6 F1, and dropping the reliability weighting 1.8 F1; both streams are necessary (the evidence-only model collapses to 48.4 AP and the claim-only model loses 5.0 AP); the ESIM four-tuple head and the retrieval-pretrained backbone each contribute about 3.6 and 4.3 AP; pseudo-gold positives beat hard positives, restricting negatives never helps, and the hard-negative count is optimal at five; the non-parametric probe on frozen embeddings collapses to 64.4 AP, while a parameter-efficient LoRA-frozen variant recovers most of the accuracy at 23% of the trainable parameters and enables gradient-free library-write adaptation. Beyond accuracy, the forward–RKC disagreement gate turns the model into a calibrated triage front-end, lifting Accuracy from 0.814 to 0.867 at 80% coverage while a random abstention does nothing.

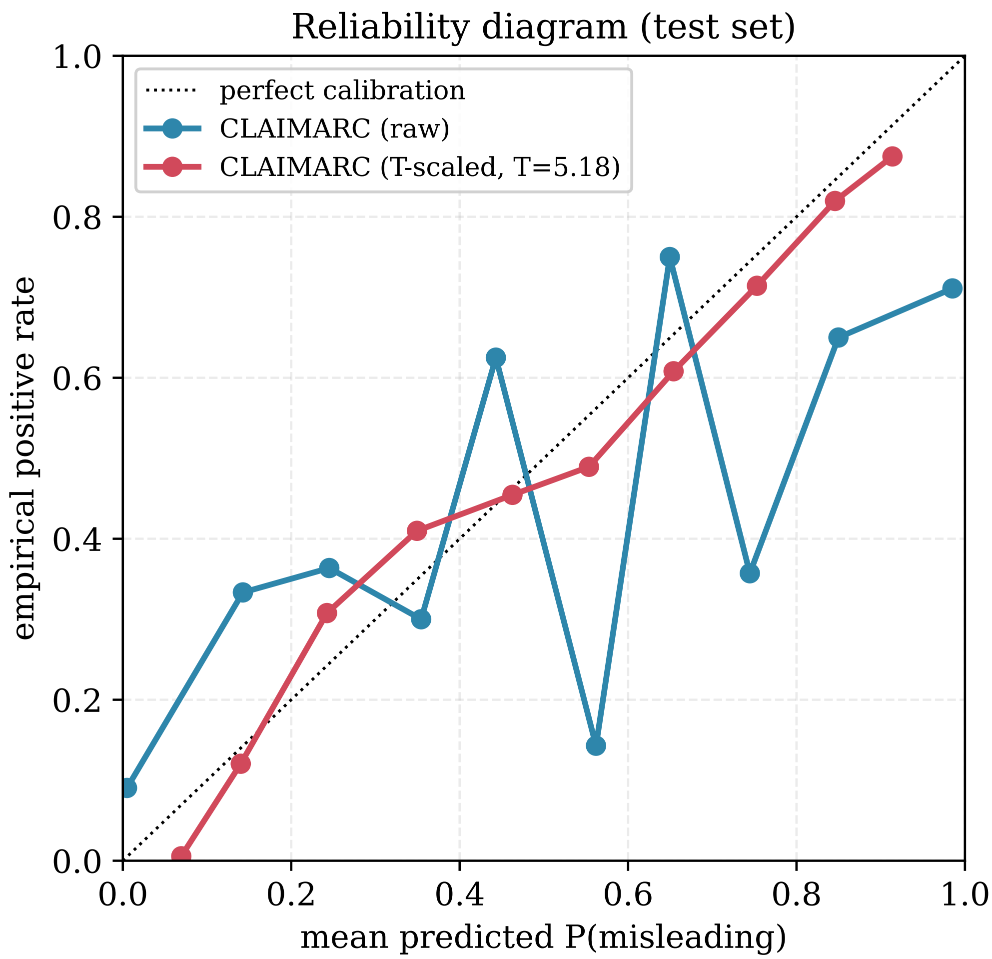

Figure 6: Reliability diagram (test). A single temperature scale brings CLAIMARC's predicted probabilities close to the diagonal.

5.2 Contributions

The study contributes on three levels. At the problem level, it redefines misleading-advertising detection in livestreaming, moving the supervision anchor from a claim-versus-knowledge-base check to an attribute-level judgment of consumer-perceived risk grounded in comments, and operationalizes that shift with a computable binary target and an instance-level reliability weight. The main comparison and the ablation together show the redefinition is empirically warranted: the fact-verification family is much weaker, and the fusion, contrast, and reliability components each carry part of the signal. At the method level, CLAIMARC folds retrieval-guided contrastive learning into joint claim–evidence discrimination, indexing the embedding space, drawing neighbors into the contrastive objective, and concentrating the learning pressure on pairs that look alike yet carry different labels; against a standard supervised contrastive baseline, the geometry analysis shows that this yields the best and uncollapsed class separation, the structure the retrieval vote relies on. At the deployment level, the framework admits a parameter-efficient variant that freezes the encoder after training and adapts through library writes, converting the absorption of new streamers, categories, and risk phrasings from model retraining into index maintenance at about a five-point AP trade-off; the cross-domain and efficiency results confirm that this pathway holds, and the disagreement gate supplies a model→human→data→model loop for compliance review. A few limits remain. The residual errors are dominated by terse endorsements with no evidential trace, where the bottleneck is signal acquisition rather than the model. This points to the next step, namely richer and cleaner evidence channels and a retrieval-augmented ensemble that combines the complementary errors of CLAIMARC and a fine-tuned encoder.

A Hyper-parameter and Variant Sweep

Table 12 reports the full single-seed sweep summarized in Section 4.9 and Figure 4. Together with the three-seed ablation battery in Tables 7–9, it spans capacity (fusion depth, heads, LoRA rank), the contrastive objective (λ_{CL}, τ, retrieval sizes, loss), architecture (feed-forward activation, cross-attention direction, projection sharing), and evidence composition. All rows read the forward classifier on the test set; the canonical configuration is the bold reference.

B Reliability-Weight Formula Sensitivity

Section 4.8 shows that the reliability weight earns its *form*; here we close the loop by perturbing the four hyper-parameters inside the formula itself: the evidence-saturation rate k, the asymmetric diagnosticity discount λ, the fake-review discount ρ, and the strong-negative bonus ϕ_{bonus}. Holding the network and training fixed, we recompute c with one parameter moved at a time and re-fit a single seed; Table 13 reads the forward classifier. Every excursion from the chosen (k=3, λ=0.3, ρ=0.4, ϕ_{bonus}=1.2) lowers AP relative to the three-seed canonical (75.4), confirming the operating point was well chosen. The two steepest sensitivities are interpretable: over-saturating the evidence count (k=6, AP 67.8) flattens the distinction between thinly and richly evidenced pairs, and weakening the strong-negative bonus (ϕ_{bonus}=1.0, AP 68.8) removes the up-weighting of pairs with explicit refuting evidence—both core to what c is meant to encode. The discount terms λ and ρ are flatter, moving AP by at most two points.

References

[1] Chen Q, Zhu X, Ling Z, Wei S, Jiang H, Inkpen D (2017) Enhanced LSTM for natural language inference. *Proc. ACL*, 1657–1668.

[2] Parikh A, Täckström O, Das D, Uszkoreit J (2016) A decomposable attention model for natural language inference. *Proc. EMNLP*, 2249–2255.

[3] Kim Y (2014) Convolutional neural networks for sentence classification. *Proc. EMNLP*, 1746–1751.

[4] Devlin J, Chang M, Lee K, Toutanova K (2019) BERT: Pre-training of deep bidirectional transformers for language understanding. *Proc. NAACL*, 4171–4186.

[5] Cui Y, Che W, Liu T, Qin B, Yang Z (2021) Pre-training with whole word masking for Chinese BERT. *IEEE/ACM Trans. Audio Speech Lang. Process.* 29:3504–3514.

[6] Xiao S, Liu Z, Zhang P, Muennighoff N (2023) C-Pack: Packed resources for general Chinese embeddings. arXiv:2309.07597.

[7] Hu E, Shen Y, Wallis P, Allen-Zhu Z, Li Y, Wang S, Wang L, Chen W (2022) LoRA: Low-rank adaptation of large language models. *Proc. ICLR*.

[8] Khosla P, Teterwak P, Wang C, et al. (2020) Supervised contrastive learning. *Proc. NeurIPS.*

[9] Wang T, Isola P (2020) Understanding contrastive representation learning through alignment and uniformity on the hypersphere. *Proc. ICML*, 9929–9939.

[10] Gao T, Yao X, Chen D (2021) SimCSE: Simple contrastive learning of sentence embeddings. *Proc. EMNLP*, 6894–6910.

[11] Mei J, Chen J, Lin W, et al. (2024) Improving hateful meme detection through retrieval-guided contrastive learning. *Proc. ACL.*

[12] Karpukhin V, Oguz B, Min S, et al. (2020) Dense passage retrieval for open-domain question answering. *Proc. EMNLP*, 6769–6781.

[13] Johnson J, Douze M, Jegou H (2019) Billion-scale similarity search with GPUs. *IEEE Trans. Big Data* 7(3):535–547.

[14] Guo C, Pleiss G, Sun Y, Weinberger K (2017) On calibration of modern neural networks. *Proc. ICML*, 1321–1330.

[15] Geifman Y, El-Yaniv R (2017) Selective classification for deep neural networks. *Proc. NeurIPS.*

[16] Li Q, Xu D J, Qian H, Wang L, Yuan M, Zeng D D (2025) A fusion pre-trained approach for identifying the cause of sarcasm remarks. *INFORMS J. Comput.* 37(2):465–479.

[17] Saito T, Rehmsmeier M (2015) The precision-recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets. *PLoS ONE* 10(3):e0118432.

Configuration	**Acc**	**F1**	**AP**	**AUC**
Canonical (N2, h8, r16, λ.5, τ.07, K35, BCE)	**80.8**	**70.5**	**70.6**	**88.6**
Fusion blocks N				
N=1	80.9	72.1	72.7	89.2
N=3	80.6	72.6	72.6	89.5
N=4	80.2	70.8	70.0	86.9
Attention heads				
4 heads	80.5	70.0	71.0	89.0
16 heads	79.9	70.9	70.5	88.9
LoRA rank				
rank 8	80.6	71.8	70.5	89.1
rank 32	81.0	73.3	73.6	89.6
Contrastive weight λ_{CL}				
0.1	79.8	70.8	69.8	88.0
0.3	79.7	70.2	69.5	87.9
1.0	80.6	71.4	70.3	88.3
Temperature τ				
0.05	79.9	70.5	69.6	88.0
0.10	79.8	70.6	69.8	88.0
0.20	80.3	70.8	69.1	88.5
Retrieval sizes (K_p, K_n)				
$(1, 1)$	79.8	70.9	70.0	88.1
$(5, 10)$	80.0	70.4	69.5	87.9
Loss function				
ASL	81.3	72.7	71.8	88.8
Focal	79.9	68.7	68.6	87.4
Architecture				
FFN GeLU (vs. SwiGLU)	80.0	69.7	71.7	88.7
cross-attn. claim→ev.	81.4	69.7	71.2	88.1
cross-attn. ev.→claim	80.3	72.4	68.0	88.3
independent projections	81.3	72.8	73.6	89.8
Evidence policy				
source-first	79.9	70.8	69.9	88.0
params only	82.0	72.4	74.1	89.8
OCR only	80.3	68.5	75.2	89.4
VLM only	81.3	73.8	75.6	90.0
multi-view mix	80.0	70.6	69.9	88.1

Table 12: Single-seed hyper-parameter and variant sweep (%, forward classifier, test set). The five core knobs (fusion depth, heads, LoRA rank, τ, (K_p, K_n)) keep AP in a 0.69–0.74 band; the canonical setting sits on the flat region. The single-source evidence policies reach high peak AP (up to 75.6) on individual views, but the canonical mix is kept for coverage and robustness when a given view is sparse.

Parameter	Setting	Acc	F1	AP	AUC
Canonical ($k3, \lambda.3, \rho.4, \phi1.2$; 3-seed)		**82.6**	**73.4**	**75.4**	**90.4**
Saturation rate k	1.5	78.7	72.4	72.6	89.3
	6.0	79.0	63.9	67.8	88.1
Asym. discount λ	0.1	81.2	73.9	71.6	89.5
	0.6	80.5	68.8	72.3	89.1
Fake-review disc. ρ	0.2	81.8	70.7	72.1	89.5
	0.6	80.9	70.4	70.0	88.8
Strong-neg bonus ϕ	1.0	78.6	72.0	68.8	88.7
	1.5	81.6	70.8	71.2	89.2

Table 13: Reliability-weight formula sensitivity (%, forward classifier, test set). One hyper-parameter of the c construction is moved at a time and c is recomputed; rows are single seed, the canonical is the three-seed reference. Every excursion lowers AP, with the saturation rate k and the strong-negative bonus ϕ_{bonus} the most sensitive.